NAME: _____ Geometry (FL B.E.S.T.)

2.2 Theorems about Parallel Lines

Once two lines are known to be parallel, every angle a transversal creates is locked into place: corresponding, alternate interior, and alternate exterior pairs are congruent, while same-side interior pairs are supplementary — never the other way around, which is a common mix-up. This section also runs the logic backward with the *converse* theorems, which let you prove lines are parallel from an angle relationship instead of assuming it. Expect multi-step algebra, a brief look at why the theorems follow from the Corresponding Angles Postulate plus Vertical Angles, and problems that chain several angle facts together.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Parallel; corresponding $5x$ and $80°$. Find x.

2. Parallel; alt. interior $x + 15$ and $55°$. Find x.

3. Parallel; same-side $3x$ and $x + 40$. Find x.

4. Parallel; alt. exterior $2x$ and $x + 28$. Find x.

5. Parallel; corresponding to $95°$?

6. Parallel; same-side to $70°$?

7. Alt. interior angles are congruent. What can you conclude about the lines?

8. Parallel; corresponding $4x - 8$ and $2x + 20$. Find x.

9. Parallel; same-side $5x + 10$ and $4x$. Find x.

10. Angle $48°$; its vertical is?

11. Angle $135°$; linear-pair partner?

12. Parallel; alt. interior $7x - 5$ and $5x + 17$. Find x.

13. Parallel; corresponding $3x + 6$ and $4x - 12$. Find x.

14. A transversal makes an angle of $(5x - 10)^\circ$ with ℓ and the corresponding angle $(3x + 24)^\circ$ with m. Find x, and give the common angle measure that guarantees $\ell \parallel m$.

15. $\ell \parallel m$. A same-side interior pair measures $4x$ and $2x + 12$. Find x, both angle measures, and the linear-pair partner of the larger angle.

✓ Answer Key

1. $x = 16$
2. $x = 40$
3. $x = 35$
4. $x = 28$
5. 95°
6. 110°
7. Parallel (Converse of the Alternate Interior Angles Theorem)
8. $x = 14$
9. $x = 20$
10. 48°
11. 45°
12. $x = 11$
13. $x = 18$
14. $x = 17$; common angle $= 75^\circ$
15. $x = 28$; angles 112° and 68°; linear-pair partner of 112° is 68°